

Air University



Computer Networks (Lab)

Year 2025

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Department of Cyber Security

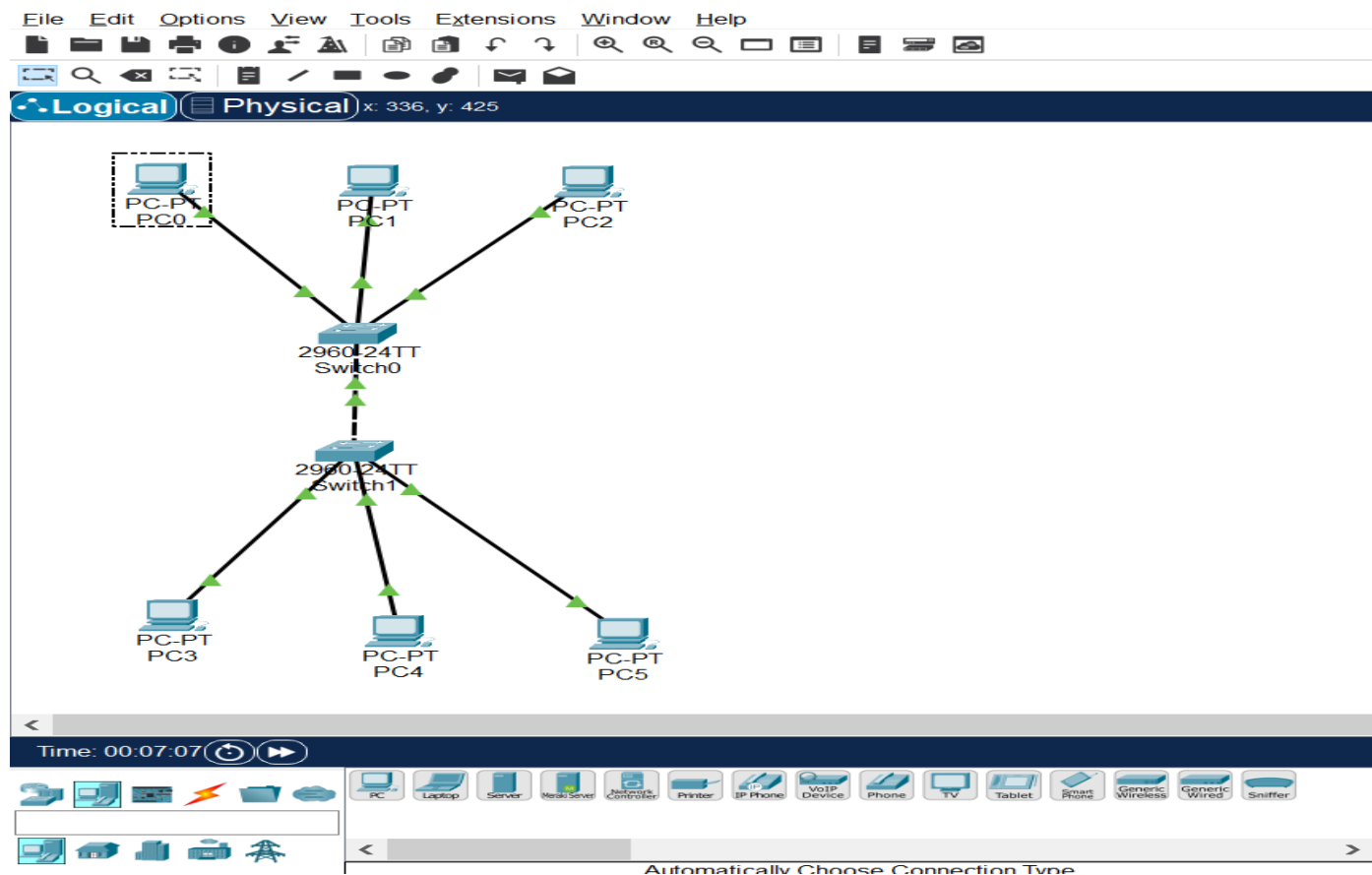
USAMA ARSHAD | 247141

BS – Cyber Security – Fall – 24 (A)

Date of Submission: October 10, 2025

Lab Task # 03 (A)

- Added required End devices and Network devices to implement given topology.



- Left click on each PC one by one, then go to Desktop and assign IP address.

The screenshot shows the configuration window for PC0 in Packet Tracer. The 'Desktop' tab is selected, and the 'IP Configuration' section is expanded. The 'Static' radio button is selected for both IPv4 and IPv6 configurations. The IPv4 Address is 192.168.7.1, Subnet Mask is 255.255.255.0, Default Gateway is 0.0.0.0, and DNS Server is 0.0.0.0. The IPv6 Address is FE80::250:FFF:FEA3:750A, Link Local Address is FE80::250:FFF:FEA3:750A, and Default Gateway is empty. The 802.1X section is also visible.

- After assigning IP, Left click on any PC and open Command Prompt from Desktop. Use the command ping along with the IP of any other PC available in the network (e.g. ping 192.168.7.5)

The screenshot displays the Cisco Packet Tracer interface. On the left, a network topology is shown with two switches, 2960-24TT Switch0 and 2960-24TT Switch1, connected by a dashed line. Switch0 is connected to three PCs (PC0, PC1, PC2) and Switch1 is connected to three PCs (PC3, PC4, PC5). The 'Physical' tab is selected. On the right, the 'PC0' configuration window is open, showing the 'Desktop' tab. A Command Prompt window is open, displaying the output of a ping command to 192.168.7.5.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.7.5

Pinging 192.168.7.5 with 32 bytes of data:

Reply from 192.168.7.5: bytes=32 time<1ms TTL=128
Reply from 192.168.7.5: bytes=32 time<1ms TTL=128
Reply from 192.168.7.5: bytes=32 time<1ms TTL=128
Reply from 192.168.7.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.7.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

- Changed the IP of PC-5 to 192.168.8.6 to check whether it stays in the network or not.

The screenshot displays the Cisco Packet Tracer interface. On the left, the same network topology is shown. On the right, the 'PC5' configuration window is open, showing the 'Desktop' tab. The 'IP Configuration' section is expanded, showing the 'Static' IP configuration for the 'FastEthernet0' interface. The IP address is set to 192.168.8.6, and the subnet mask is 255.255.255.0. The 'IPv6 Configuration' section is also expanded, showing the 'Static' configuration with a Link Local Address of FE80::230:A3FF:FE3B:B767.

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.8.6

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::230:A3FF:FE3B:B767

Default Gateway:

DNS Server:

802.1X

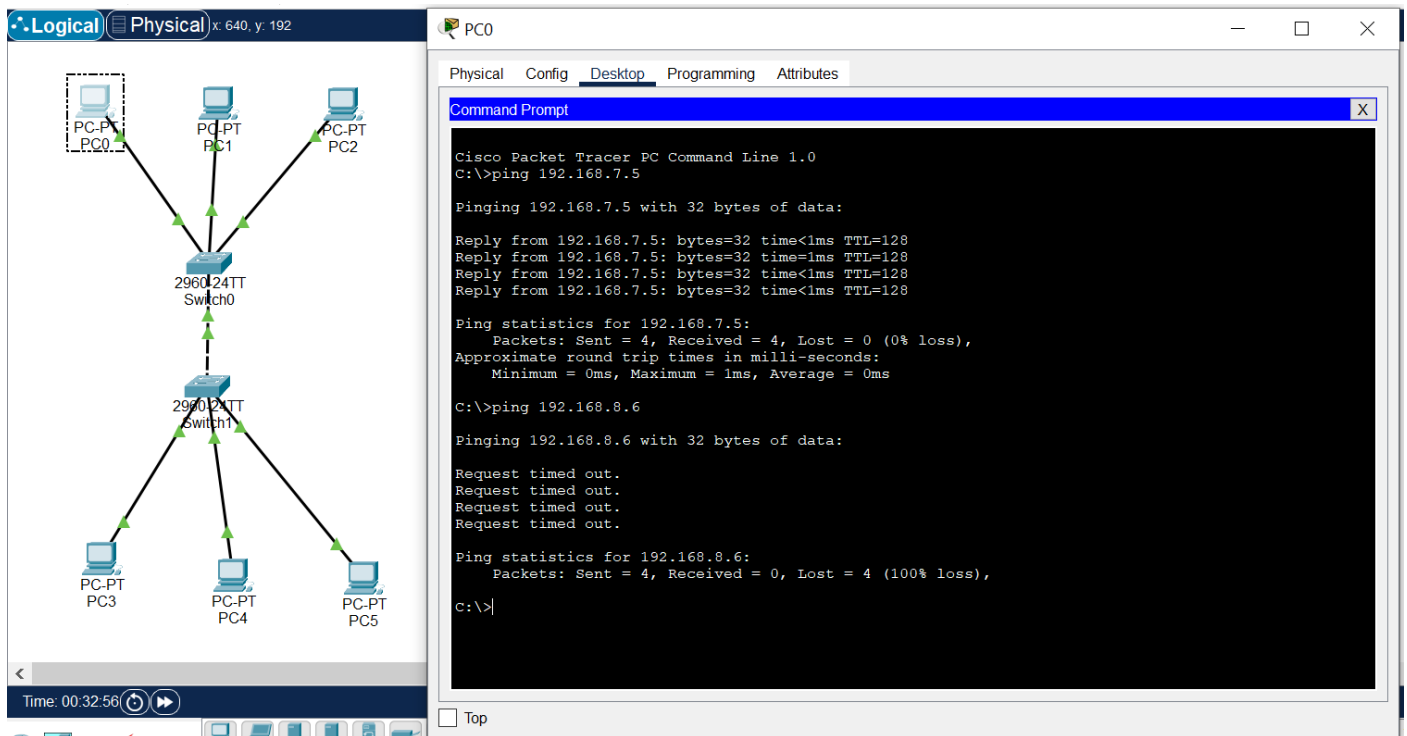
☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

- Checking the PC-5 from PC-0 using ping command, we can see that PC-5 does not exist in the network.



The screenshot shows a network topology in Cisco Packet Tracer. On the left, a network diagram displays two switches, 2960-24TT Switch0 and 2960-24TT Switch1, connected to each other. Switch0 is connected to three PCs: PC-PT PC0, PC-PT PC1, and PC-PT PC2. Switch1 is connected to three PCs: PC-PT PC3, PC-PT PC4, and PC-PT PC5. On the right, a PC window for PC0 shows a Command Prompt with the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.7.5

Pinging 192.168.7.5 with 32 bytes of data:

Reply from 192.168.7.5: bytes=32 time<1ms TTL=128
Reply from 192.168.7.5: bytes=32 time=1ms TTL=128
Reply from 192.168.7.5: bytes=32 time<1ms TTL=128
Reply from 192.168.7.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.7.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.8.6

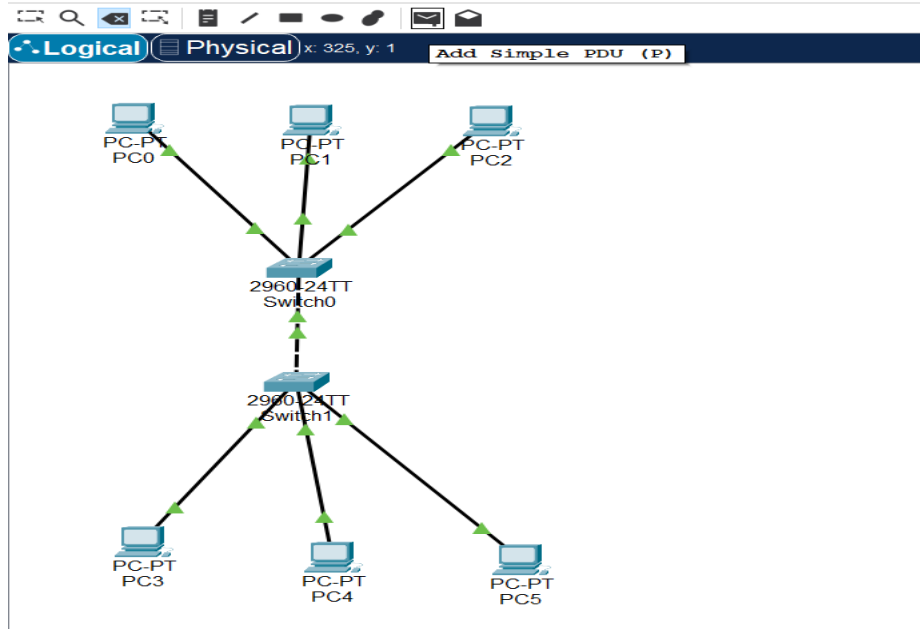
Pinging 192.168.8.6 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.8.6:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

- Add simple PDU for sending packets from one pc to another.



- The packet has successfully transferred.

Realtime Simulation										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC4	ICMP		0.000	N	0	(edit)	(delete)

- Run the simulation to see the packets moving from source to destination.

The screenshot shows the Cisco Packet Tracer interface. The main workspace displays a network topology with two switches (Switch0 and Switch1) and six PCs (PC0-PC5). Switch0 is connected to PC0, PC1, and PC2. Switch1 is connected to PC3, PC4, and PC5. The Simulation Panel on the right shows a list of events, including DTP and STP messages between switches and PCs. The bottom status bar shows the simulation is running in Realtime mode.

- Left click on any switch and then go to CLI and run the command “show mac address” to see the connected devices.

The screenshot shows the Cisco Packet Tracer interface with the CLI of Switch1 open. The CLI window displays the output of the 'show mac address' command, showing a table of MAC addresses for VLAN 1.

Vlan	Mac Address	Type	Ports
1	000a.41e9.14ac	DYNAMIC	Fa0/4
1	00d0.ff29.4a04	DYNAMIC	Fa0/4
1	00e0.b0d2.828b	DYNAMIC	Fa0/2

THE END